IZTECH- Electronics and Communications Engineering
EE451 – Communication II

1. Modulation and demodulation of BFSK, QPSK and 16-QAM with BER analysis and MATLAB GUI

2. Team Members

Name - Surname	Student ID
Halil Yılmaz	300206051
Emir Dündar	310206021
Atlas Ferhat Hacımusalar	310206038

3. Abstract

Digital communication systems must balance data rate, bandwidth and robustness to noise when choosing a modulation scheme. This project compares Binary Frequency Shift Keying (BFSK), Quadrature Phase Shift Keying (QPSK) and 16-ary Quadrature Amplitude Modulation (16-QAM) by their Bit Error Rate (BER), both in simulation and over a real radio link.

We will implement each modulator and demodulator from first principles in MATLAB. First, signals pass through a simulated AWGN channel and BER is measured against Eb/N0 by Monte Carlo simulation, validated against closed-form theory. Second, the same signals are sent between two computers, each connected to an ADALM-Pluto software-defined radio; the receiving node performs frequency and timing synchronization, demodulates and computes BER. Both nodes can transmit and receive, so the link works in both directions. A MATLAB GUI on the receiving computer selects the modulation and link settings and shows the received constellation, the waveform and the measured BER against simulated and theoretical curves.

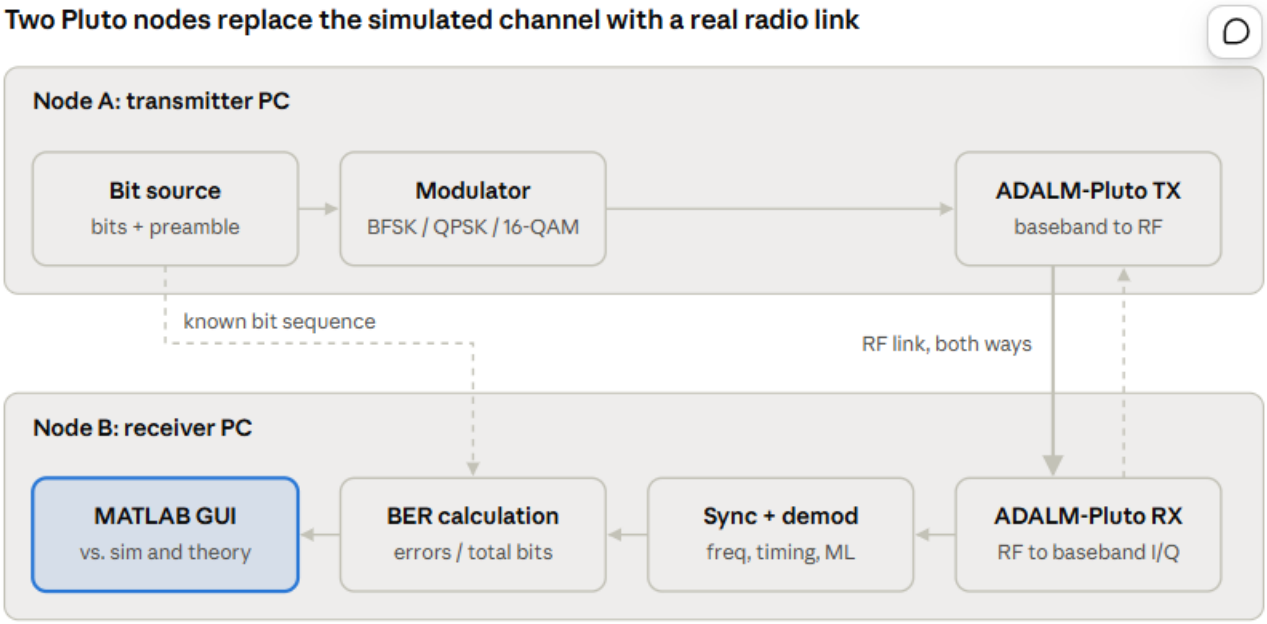
Expected outcomes are working implementations of all three schemes, simulated BER curves that match theory, a two-way Pluto link, and a comparison of measured and simulated BER that quantifies the power–bandwidth trade-off and the effect of real-channel impairments.

Tools: MATLAB (Communications Toolbox, Signal Processing Toolbox, App Designer), Communications Toolbox Support Package for ADALM-Pluto Radio, two ADALM-Pluto SDRs and two computers.

4. Problem Definition & Motivation

Every wireless system, from Wi-Fi to cellular networks, must decide how many bits to place on each transmitted symbol. Higher-order schemes such as 16-QAM increase the data rate in a fixed bandwidth, but their constellation points are closer together, so noise causes more bit errors. Lower-order schemes such as BFSK are robust but waste bandwidth. The problem is to quantify this trade-off: how much extra Eb/N0 does each scheme need to reach a target BER, and what does it gain in return? Answering this with our own implementations, validated against theory and then measured over a real ADALM-Pluto radio link, shows both the ideal trade-off and how real hardware departs from it.

5. Block Diagram of the Proposed System



Node A generates a known bit sequence with a preamble, modulates it and transmits it through its ADALM-Pluto. Node B receives the RF signal with a second Pluto, corrects frequency and timing offsets, demodulates, and compares the result with the known sequence to compute BER; the GUI plots it against the simulated AWGN and theoretical curves. The roles can be swapped, so each node both transmits and receives.

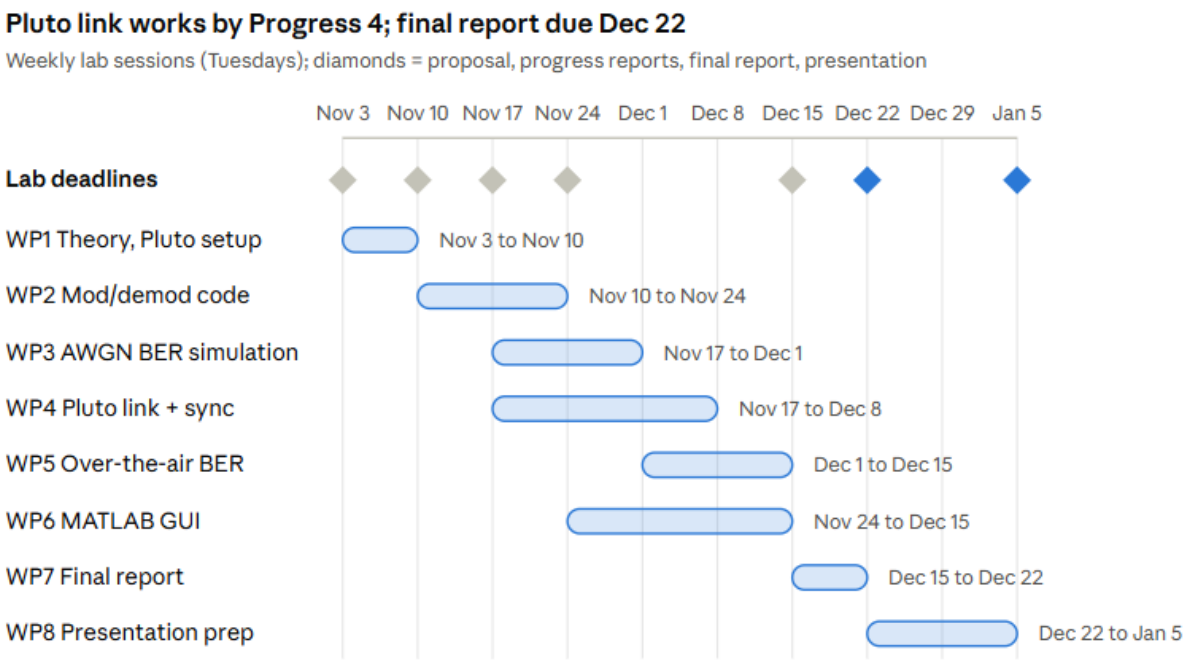
6. Light Literature Awareness

Proakis & Salehi, Digital Communications. The standard textbook treatment of digital modulation over AWGN channels. It derives the optimum (maximum-likelihood) detector and the closed-form BER expressions for FSK, PSK and QAM. We will use these expressions as the theoretical reference that our simulated curves must match.

Haykin, Communication Systems. Presents signal-space representation and coherent versus noncoherent detection of binary FSK. Relevant for deciding which BFSK receiver we implement, since the two have different theoretical BER curves.

MathWorks, Communications Toolbox and ADALM-Pluto Support Package documentation. Covers built-in modulators, the AWGN channel and BER tools, plus QPSK transmitter/receiver examples on ADALM-Pluto with carrier and timing synchronization. We will use the toolbox functions to verify our own code, and the Pluto examples as the starting point for our over-the-air link.

7. Project Timeline (Gantt Chart)



8. References

1. J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. New York: McGraw-Hill, 2008.
2. S. Haykin, *Communication Systems*, 5th ed. Hoboken, NJ: Wiley, 2009.
3. MathWorks, *Communications Toolbox Support Package for Analog Devices ADALM-Pluto Radio* — documentation and examples.